DATA BASES – PROF. D. M. BRAGA – A. A. 2021/22

CALL 5 – AUGUST 29TH 2022 – DURATION: 1h 45m

SQL (24 points)

The schema describes the database of a car insurance company. Please note that a crash may involve one, two or many cars. For simplicity, all cars have one owner.



CAR (PlateNo, PolicyID, OwnerID, Make, Model, Year)
CLIENT(OwnerID, BirthDate, CityOfResidence, Nationality)
ACCIDENT (AccID, Date, Time, City, Address)
INVOLVEMENT (AccID, PlateNo, IncurredDamage, CausedDamage)

1. Briefly but precisely explain what is extracted by the following view and query [**5 p**]:

a.
```
create view acc( OwnId, Year, Num, CsdDmg ) as
    select OwnerId, Date.year(),
           count( * ), sum( CausedDamage )
    from CAR natural join INVOLVEMENT
         natural join ACCIDENT
    group by OwnerId, Date.year()
```

b.
```
select AccID, Date, count( * )
from ACCIDENT A natural join INVOLVEMENT
group by AccID
having count( * ) >= ALL ( select count( * )
                           from ACCIDENT natural join INVOLVEMENT
                           where Date = A.Date
                           group by AccID )
```

2. Extract in SQL:

- **a**. The owners who never had accidents in their home city; [**5 p**]
- **c**. The owners, involved in 3 or more accidents, whose accidents all and always took place on their birthday. [**5 p**]
- **b**. The average number of cars involved in the accidents of each city (*like*: <Chicago, 2.4>, <Detroit, 1.7>, <Baltimore, 2.7>... *pay attention to correctly count the accidents and the cars, compute the ratio, and the average*); [**5 p**]

3. Assume that the view defined in **1.a** is materialized. Formulate a set of triggers that incrementally maintain the materialization of the view in response to *insertions* on table INVOLVEMENT (all other events can be neglected, and you can assume that the tuple in *Accident* is always already present whenever an *Involvement* is inserted) [**4 p**]

Conceptual Design (8 points)

4. Design an E-R model for the database described in the following scenario:

A used motorbike dealers chain has several sale points in the country. Each agency has a unique code, a name, an address, a phone number and an email address. Name and SSN are also stored for the General Manager of each agency and for the (possibly other) employees, together with their hire dates.

Each sale point sells its own motorbikes, physically positioned in slots that are identified by a progressive number (that is unique in each sale point) and can be inside or outside the building; the actual current position of each bike is recorded. Bikes are stored with the following data: plate number, make, model, color, displacement, year of manufacturing and mileage. Bikes can be fuel-propelled or electric: for the latter ones we also store the autonomy (expressed in Kms) for the different types of usage, in city traffic, mixed and on highways. Whenever a bike is purchased by a customer, data are stored for the transaction: date, price and SSN, name birthdate and address of the buyer.

- *The project is evaluated for completeness, correctness, legibility, minimality and self-explanatory nature.*
- *Remember to specify the **identifiers** of all entities and the **cardinalities** of all associations.*